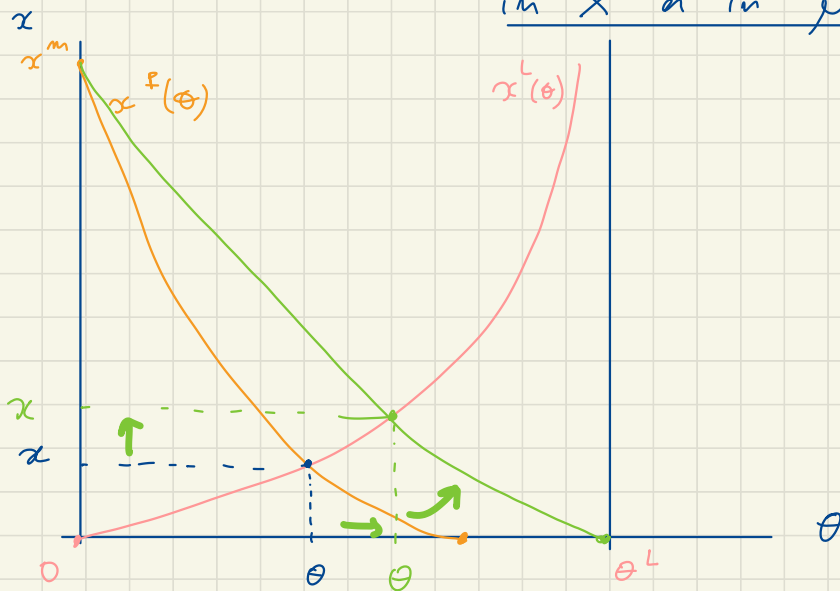


Aggregate Demand Shocks with Fixed Prices

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Aggregate demand shock:

Positive shock: increase
in X or in μ



$$x^f(\theta) = \tau^{-1} \left(\left[\frac{x^L \mu \alpha}{w \cdot L} \cdot \frac{1}{f(\theta)} \right]^{\frac{1}{\varepsilon-1}} - 1 \right)$$

increasing

$$x^l(\theta) = f^{-1} \left(\frac{w/p}{\alpha \alpha} h^{1-\alpha} f(\theta)^{1-\alpha} [1 + \tau(\theta)]^\alpha \right)$$

After an $\uparrow$ in AD ($\uparrow$ in X or $\uparrow$ in μ).

- $\theta \uparrow$ $[\hat{f}(\theta) \uparrow, q(\theta) \downarrow, \hat{\tau}(\theta) \uparrow]$
- $x \uparrow$ $[f(x) \uparrow, q(x) \downarrow, \tau(x) \uparrow]$
- $L = \hat{f}(\theta) \cdot h$ so $L \uparrow$
- what happens to η ? unclear from AD curve $\rightarrow$ need a $\pm$ approach $\eta \uparrow$.

$$\left[y = \frac{\uparrow X^{\frac{1}{\epsilon}}}{[1 + \tau(x)]^{\frac{1}{\epsilon-1}}} \cdot \frac{\mu^{\frac{1}{\epsilon}}}{\rho} \right]$$

$$(L) \quad f(x) = \frac{w/p}{a \cdot l} \cdot l^{1-\alpha} \cdot \hat{f}(\theta)^{1-\alpha} [1 + \tau(\theta)]^{\alpha}$$

$$\Rightarrow \alpha \cdot f(x) \cdot a \cdot \underbrace{\left[\frac{\hat{f}(\theta) \cdot l}{1 + \tau(\theta)} \right]^{\alpha}}_n = (w/p) \cdot \underbrace{\hat{f}(\theta) \cdot l}_l$$

$f(x) \cdot l = y$

$$\Rightarrow \alpha \cdot y = (w/p) \cdot l^{\uparrow}$$

labor share = α

↓
labor income
total income

From this we see $y \uparrow$

$$= \frac{w \cdot l}{p \cdot y} = \alpha$$

- unemployment rate $1 - \hat{f}(\theta) \quad \downarrow$

- idleness rate $1 - f(x) \quad \downarrow$

$\Rightarrow$ less slack